

GCP Hands-On:

Hands-on experience + documentation of core GCP features

Buckets:

Pre-Requisite Steps:

Select a project New project

Search projects and folders

Recent Starred All

Name	Type	ID	
✓ Portfolio ?	Project	portfolio-490106	☆
wazuh-dashboard ?	Project	wazuh-dashboard-484512	☆

Google Cloud Search (/) for resources, docs, products and more

New Project

You have 14 projects remaining in your quota. Request an increase or delete projects. [Learn more](#) ?

[Manage Quotas](#) ?

Project name *
GCP Hands On ?

Project ID: gcp-hands-on-490407. It cannot be changed later. [Edit](#)

Parent resource *
 No organisation [Browse](#)

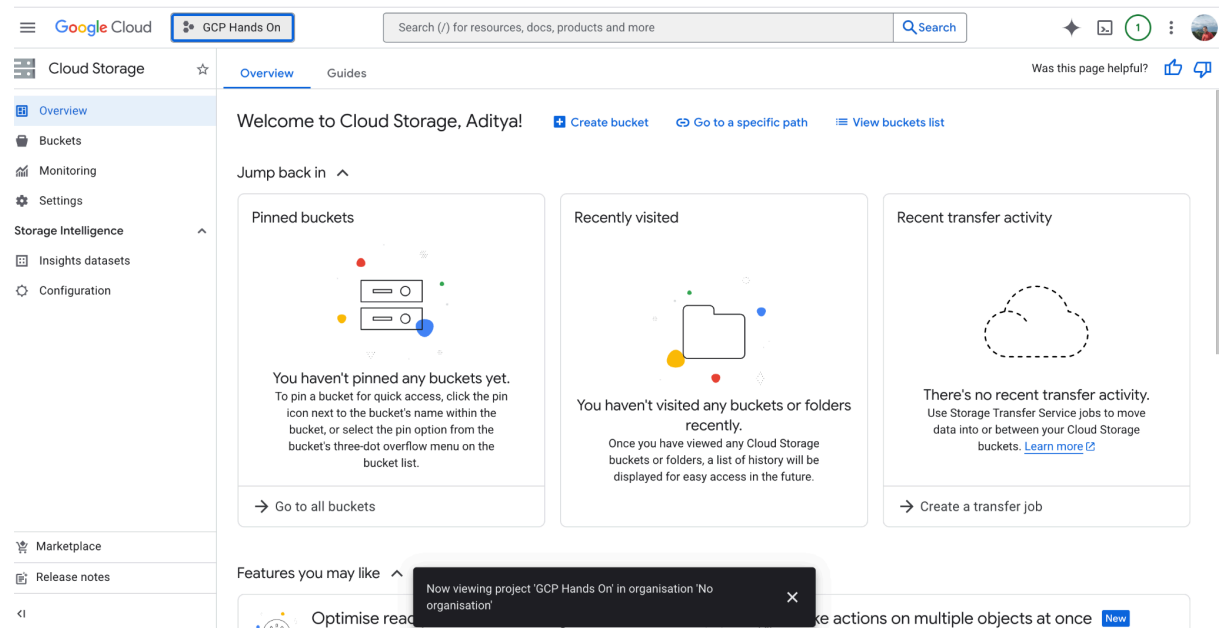
Select the organisation or folder where this resource will be created

I've created a new project named GCP Hands On

Name	Type	ID	
GCP Hands On ?	Project	gcp-hands-on-490407	☆
✓ Portfolio ?	Project	portfolio-490106	☆

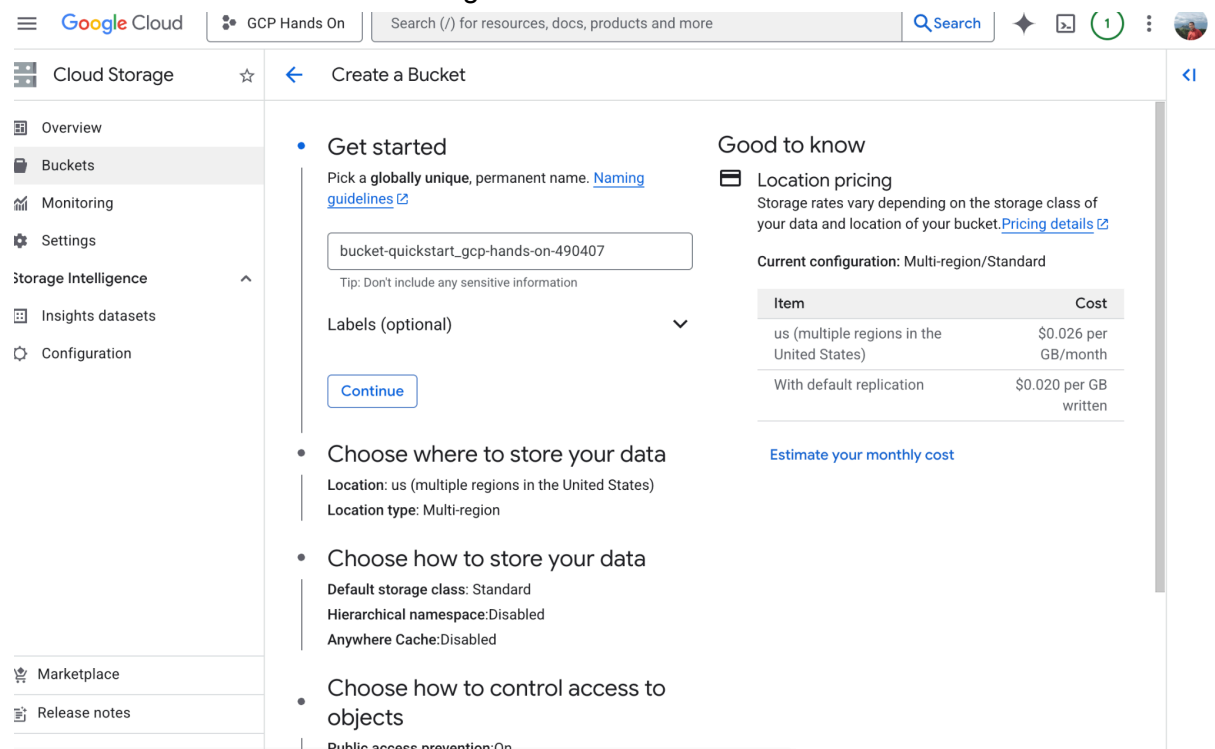
The Portfolio project is still ticked because I was navigating from that project's root. I'll navigate to the newly created project. Now as you can see we're in the GCP Hands On

project.



Bucket Creation:

Now I'll head over to the buckets section in the navbar and create a bucket. This helps me store data in buckets in cloud storage.



Choose the location nearest to you and is stable/available:

Note: I've chosen Singapore although Mumbai is closer to me because of availability issues in the Indian, Middle East, and European servers. Eastern Asia (Tokyo, Seoul, etc.) have a longer RTT, so among the stable options I have, Singapore is the closest geographically.

- ## Choose where to store your data

This choice defines the geographic placement of your data and affects cost, performance and availability. Cannot be changed later. [Learn more](#) 

Location type

- ☐ **Multi-region**
Highest availability across largest area
- ☐ **Dual-region**
High availability and low latency across 2 regions
- ☒ **Region**
Lowest latency within a single region

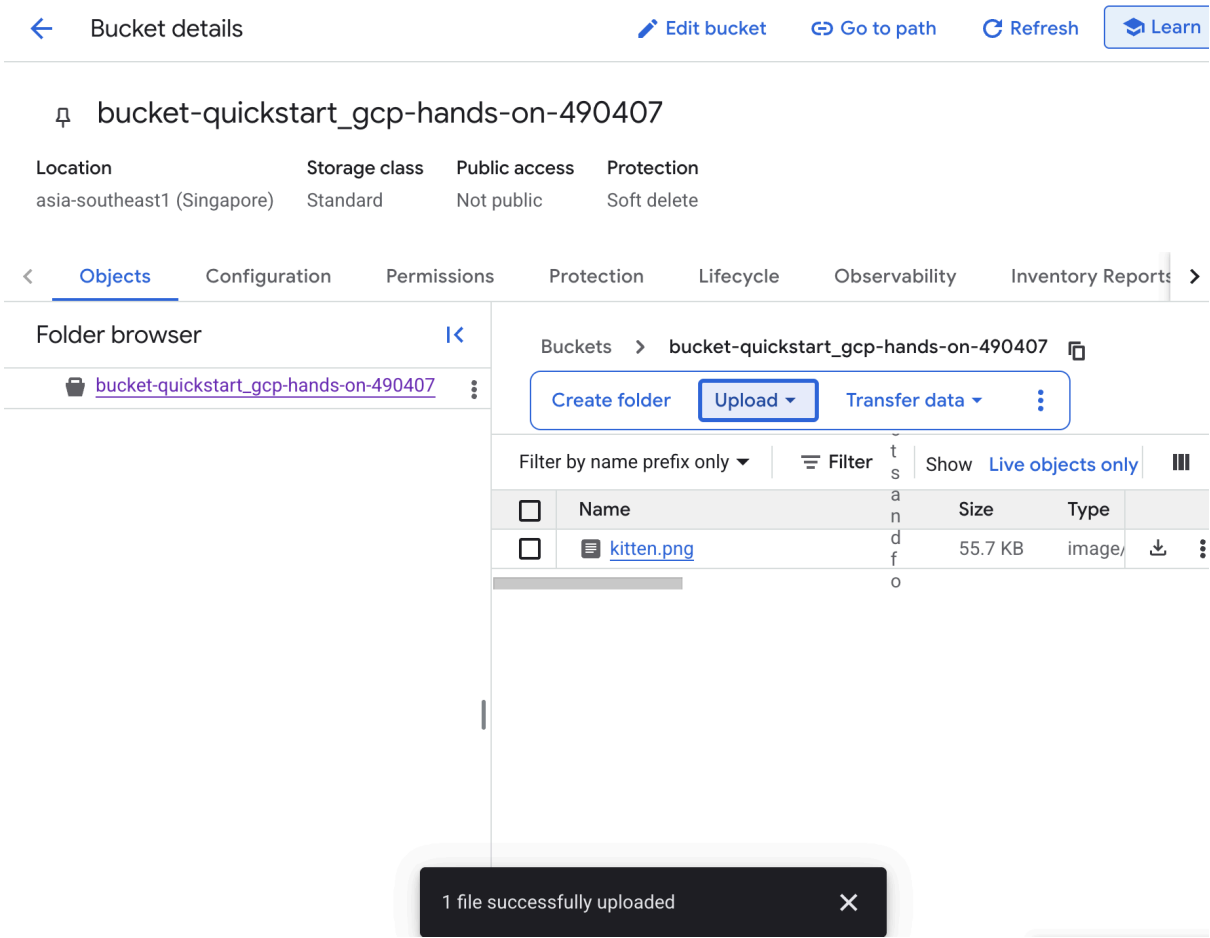
asia-southeast1 (Singapore) ▼

- ☐ **Add cross-bucket replication via Storage Transfer Service**
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings – e.g. region, storage class, etc. [Learn more](#) 

[Continue](#)

Data Upload & File Organization:

I've successfully uploaded an image of a kitten, named kitten.png using the upload button.



Cloud Storage

Overview

Buckets

Monitoring

Settings

storage Intelligence

Insights datasets

Configuration

Object details

Buckets

bucket-quickstart_gcp-hands-on-490407

kitten.png

Live object

Version history

Download

Edit metadata

Edit access

Delete

Overview

Type	image/png
Size	55.7 KB
Created	16 Mar 2026, 13:04:39
Last modified	16 Mar 2026, 13:04:39
Storage class	Standard
Custom time	—
Public URL	https://storage.googleapis.com/bucket-quickstart_gcp-hands-on-490407/kitten.png
Authenticated URL	https://storage.cloud.google.com/bucket-quickstart_gcp-hands-on-490407/kitten.png
gsutil URI	gs://bucket-quickstart_gcp-hands-on-490407/kitten.png

Permissions

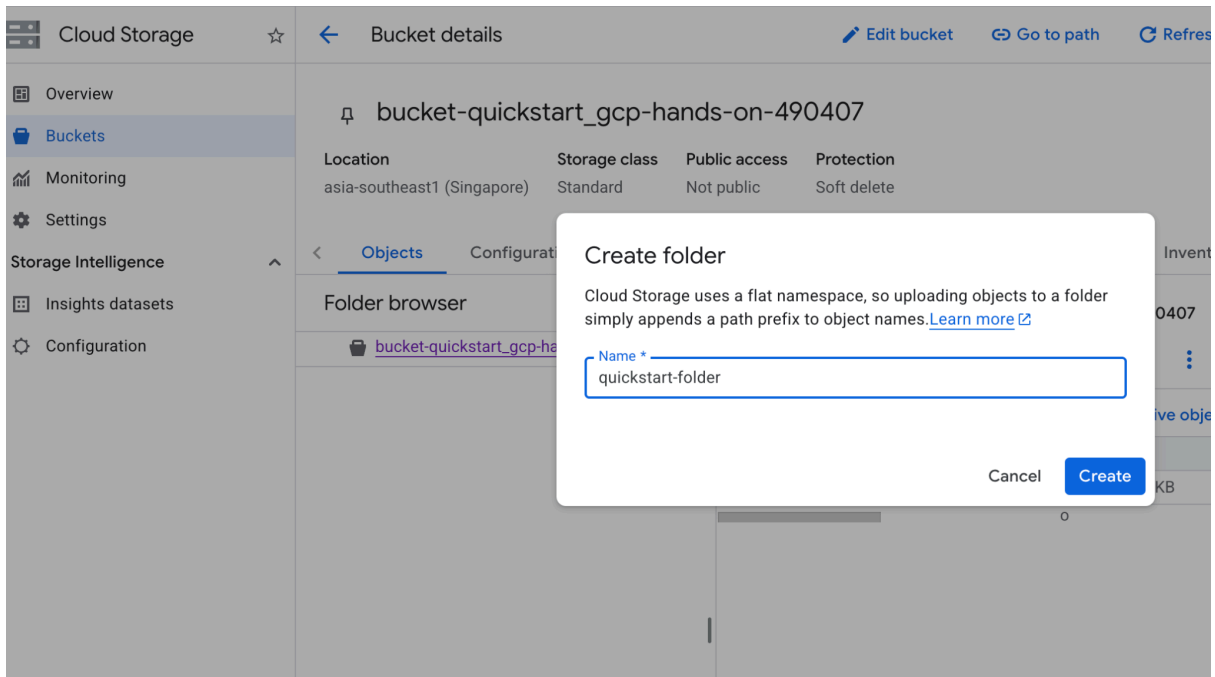
Public access	Not public
---------------	------------

Protection

Version history	—
Retention expiry time	None
Object retention retain-until time	None
Bucket retention retain-until time	None
Hold status	None
Encryption type	Google-managed

Now to move the image into a dedicated folder, I'll first create a folder using the create folder button, and then click on the three vertical dots next to the kitten.png file and click move, and then browse and select my destination folder and move it.

I've opened the gsutil equivalent snippet if you're curious to try this on a terminal.



Move object

Source object path

bucket-quickstart_gcp-hands-on-490407/kitten.png

Destination * ☒

- ☐ Keep source permissions ☒ Use default permissions at destination

```
$ gsutil mv gs://bucket-quickstart_gcp-hands-on-490407/kitten.png 
```

[^ gsutil equivalent](#)

bucket-quickstart_gcp-hands-on-490407

Location	Storage class	Public access	Protection
asia-southeast1 (Singapore)	Standard	Not public	Soft delete

[Objects](#) [Configuration](#) [Permissions](#) [Protection](#) [Lifecycle](#) [Observability](#) [Inventory Reports](#) >

Folder browser

bucket-quickstart_gcp-hands-on-490407

quickstart-folder/

Create folder

Other services

Upload

by name prefix only

Filter

Show Live objects only

Name	Size	Type
kitten.png	55.7 KB	image/

Moved object to bucket-quickstart_gcp-hands-on-490407/quickstart-f... [Go to object](#) X

When you make Cloud Storage buckets publicly readable, anyone on the internet can list and view their objects and view their metadata (excluding ACLs). *Don't include sensitive info here!*

Access Management:

GCP Web GUI:

Grant access to bucket-quickstart_gcp-hands-on-490407

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

 bucket-quickstart_gcp-hands-on-490407

Add principals

Principals are users, groups, domains or service accounts. [Learn more about principals in IAM](#)



Adding 'allUsers' makes this resource **public and accessible to anyone on the Internet**.

New principals *

allUsers 



Principals allUsers and allAuthenticatedUsers cannot be added since public access prevention is enforced on this bucket.

Assign Roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *

Storage Object Viewer 

IAM condition (optional) 

[+ Add IAM condition](#)



Grants access to view objects and their metadata, excluding ACLs. Can also list the objects in a bucket.

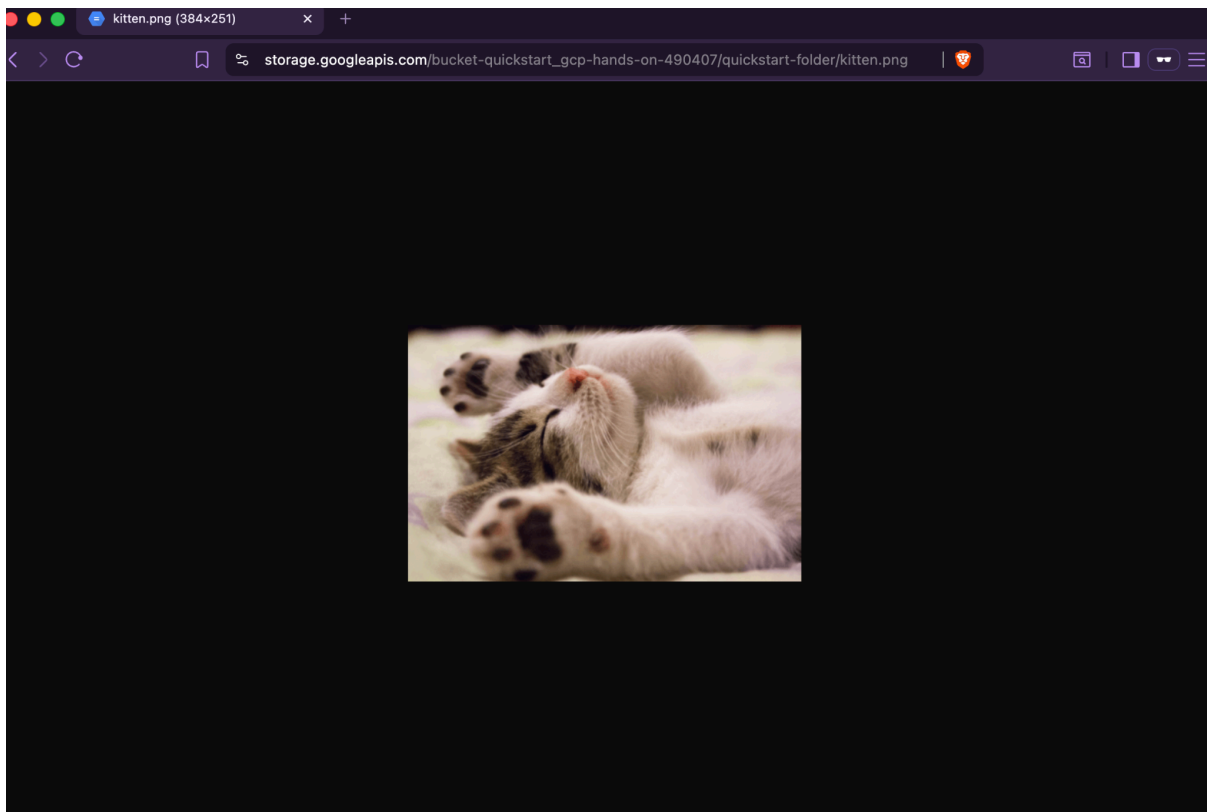
[+ Add another role](#)

I am trying to grant access to everyone i.e. the public by listing allUsers but as you can see it is throwing an exception, because I had enabled public access prevention while creating this bucket. Since this project is not linked to any organization, the bucket settings page is slightly different as compared to what would've been the case otherwise. Hence, the obvious next step is to utilize the terminal that GCP provides built in.

GCloud CLI:


```
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud storage buckets add-iam-policy-binding gs://bucket-quickstart_gcp-hands-on-490407 --member="allUsers" --role="roles/storage.objectViewer"
bindings:
- members:
  - projectEditor:gcp-hands-on-490407
  - projectOwner:gcp-hands-on-490407
  role: roles/storage.legacyBucketOwner
- members:
  - projectViewer:gcp-hands-on-490407
  role: roles/storage.legacyBucketReader
- members:
  - projectEditor:gcp-hands-on-490407
  - projectOwner:gcp-hands-on-490407
  role: roles/storage.legacyObjectOwner
- members:
  - projectViewer:gcp-hands-on-490407
  role: roles/storage.legacyObjectReader
- members:
  - allUsers
  role: roles/storage.objectViewer
etag: CAM=
kind: storage#policy
resourceId: projects/_/buckets/bucket-quickstart_gcp-hands-on-490407
version: 1
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$
```

As you can see, allUsers is granted viewing permission regardless of what the GUI tells. The GUI may still show the old issue, this is a normal cache related issue. Below you can see the kitten.png opened using the public link a private tab in a different browser. This means the terminal fix has worked perfectly!



Since this hands-on exercise is done, I've deleted the folder. Note that these will be in a soft delete state, that is, not permanently deleted until next week.

But to prevent recurring charges, it is **IMPORTANT** to delete the bucket itself if there's no more files/folders to be added in this bucket.

Deletion & Clean-Up:

Delete bucket bucket-quickstart_gcp-hands-on-490407?



This action will **soft delete the bucket, the objects that it contains (including versions) and any associated Anywhere Caches** from Google Cloud. After the soft delete retention period of 7 days, data will be permanently deleted and not recoverable.

You are deleting the bucket **bucket-quickstart_gcp-hands-on-490407**.

Deletion is performed in the background and may take some time depending on the size of the objects being deleted.

Confirm deletion by typing DELETE below:

Cancel

Delete

Conclusion:

This hands-on exercise served as a deep dive into the practicalities of **Google Cloud Storage (GCS)**, **Identity & Access Management (IAM)**, and the **GCloud CLI**. It moved beyond theoretical knowledge to solve real-world configuration challenges.

Key Takeaways

- **Cloud Architecture & Latency Optimization:** Strategically selected the **asia-southeast1** (Singapore) region over closer alternatives. This decision was based on a calculated trade-off between physical distance (RTT) and regional infrastructure stability, demonstrating an SRE-mindset toward service availability.
- **CLI vs. GUI (Source of Truth):** Encountered a critical discrepancy where the GCP Web Console (GUI) enforced "Public Access Prevention" due to stale state/caching. Successfully bypassed this by utilizing the **GCloud CLI** to modify IAM policy bindings directly. This reinforced the principle that the **API is the source of truth** in GCP and other cloud environments.

- **Security & IAM Governance:** Practiced the principle of public exposure management by navigating the transition from a private, "Enforced" bucket state to a public-serving state using the `allUsers` principal. Gained a clear understanding of how Organization-level policies impact project-level permissions.
- **Resource Lifecycle Management:** Implemented proactive cleanup by utilizing `gsutil` and console commands to delete objects and buckets. Recognized the implications of **Soft Delete** policies and the importance of resource termination to prevent "cloud bill sprawl."